
UK CAA EASA PPL REVISION PAPERS

Practical Navigation – Practice Planning and Plogs 125 - 165

Blank Answer Sheets 167 - 170

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PRACTICAL NAVIGATION

PRACTICE PLANNING AND PLOGS

The following routes are provided to give practice in the practical application of the navigation planning techniques that you have learned and will put to use when cross country flying. **This section does not form part of the theoretical examination preparation, you will not be required to plot and plan a whole route. It is included to help you hone your navigation planning skills. Questions relating to each route are also provided to assist you in becoming more familiar with the chart and the symbols used. Chart symbols are in the syllabus for the CAA examination, this is a good way to learn some of them and what actions they should prompt from a pilot.**

NAVIGATION ROUTE 1

You are planning the following VFR flight:

From: Leicester (52 36 28N 001 01 55W)

To: Thruxton (51 12 38N 001 36 00W)

Via: Westcott NDB (WCO 51 51 18N 000 57 75W)

Alternate: Popham (51 11 63N 001 14 08W)

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
LEICESTER	WCO													
52 36 28N	51 51 18N	As Reqd.	1300	110	150/25				3W					
001 01 55W	000 57 75W													
WCO	THRUXTON													
51 51 18N	51 12 38N	As Reqd.	1300	110	150/25				3W					
000 57 75W	001 36 00W													
											TOTAL			
ALTERNATE														
THRUXTON	POPHAM													
51 12 38N	51 11 63N	As Reqd.	1300	110	160/20				3W					
001 36 00W	001 14 08W													

Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet, within 5 nm of track rounded up to the next 100 feet

ANSWER: Completed Plog.

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
LEICESTER	WCO													
52 36 28N	51 51 18N	As Req.	2300	110	150/25	178	6S	172	3W	175	87	45	31	
001 01 55W	000 57 75W													
WCO	THRUXTON													
51 51 18N	51 12 38N	As Req.	2300	110	150/25	213	12S	201	3W	204	96	45	28	
000 57 75W	001 36 00W													
											TOTAL	90	59	
ALTERNATE														
THRUXTON	POPHAM													
51 12 38N	51 11 63N	As Req.	1800	110	160/20	095	10P	105	3W	108	100	13.7	8	
001 36 00W	001 14 08W													

Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet, within 5 nm of track rounded up to the next 100 feet

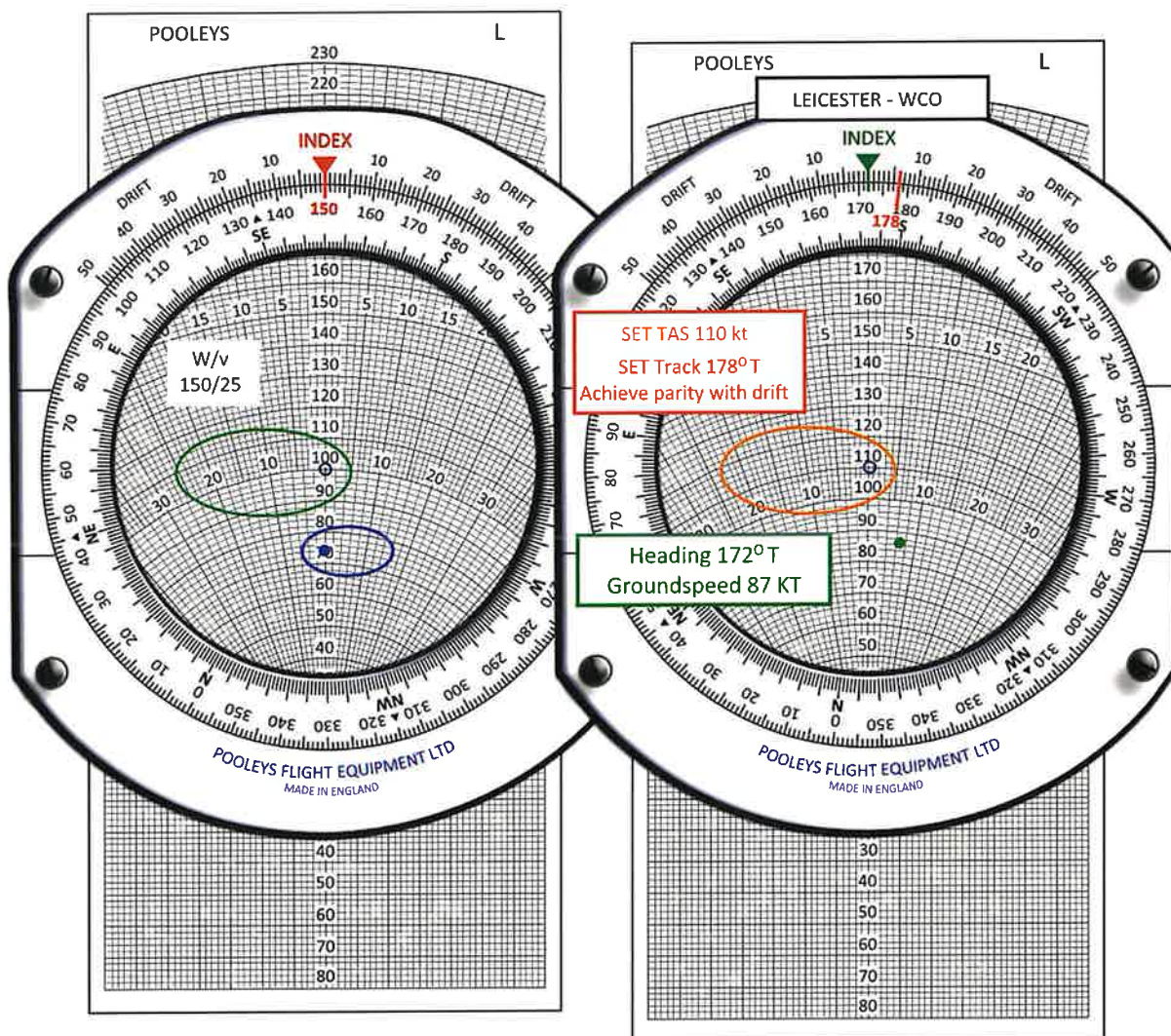
Method: Legs Leicester to WCO; WCO to Thruxton:

1. Place the centre of the protractor over the track drawn on the map near the mid-point between the two ends of the track.
2. Align the grid marks on the protractor with the vertical meridian lines on the chart. Read off the True Track, in this case 178° for the first leg and 213° from WCO to Thruxton.
3. Next, find the effect of the wind on the aircraft. The wind vector is the same for both legs, so using your CRP computer set the wind direction (150) on the rotating inner scale under the Index on the fixed outer scale.
4. Set the centre circle on a convenient speed arc – 100 is a handy round number. Mark down the wind speed, 25 knots for this question.
5. Move the sliding scale so that the centre circle is over the TAS 110.

Leicester to WCO:

6. Rotate the inner scale so that the measured true track, 178, is now below the index. Note the wind mark is now displaced 8° to the right.
7. Rotate the inner scale 8° to the right, 178 is now below 8° on the right hand side of the fixed drift scale. However, now the pencil mark indicates 6° of starboard drift. What we are aiming for is parity, so move the rotating scale so that 178 is below 6° on the fixed drift scale and still right of the index. In so doing the drift remains at 6° and we have parity – the track is offset 6° and the wind mark still shows 6°.

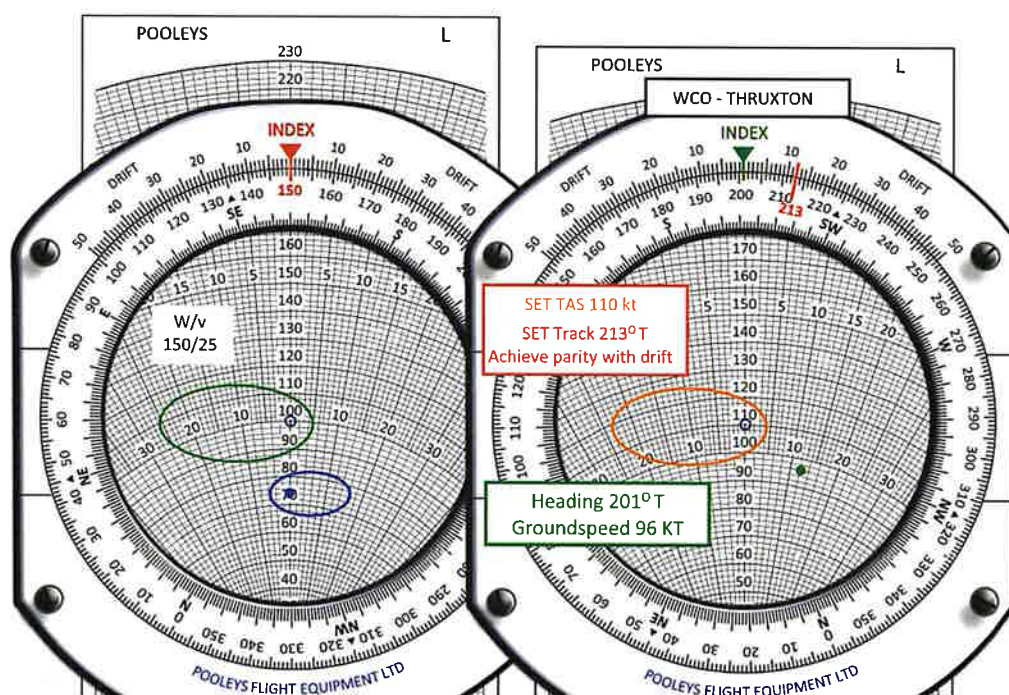
8. Read off the true heading under the Index = 172° .
9. Finally apply the variation, in this case 3° W. Remember "East is Least, West is Best", the variation is West, hence we add it giving a magnetic heading of 175° .
10. Read the ground speed under the wind dot – 87 knots.



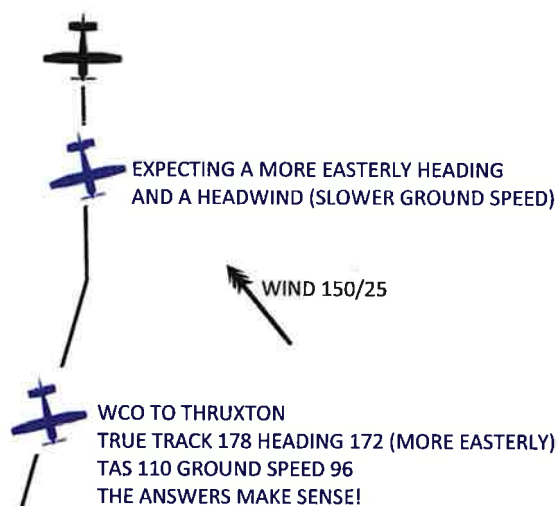
WCO to Thruxton:

As the wind is the same for the second leg, rotate the inner scale so that the measured true track, 213, is now below the index. You can repeat stages 6 to 10 above.

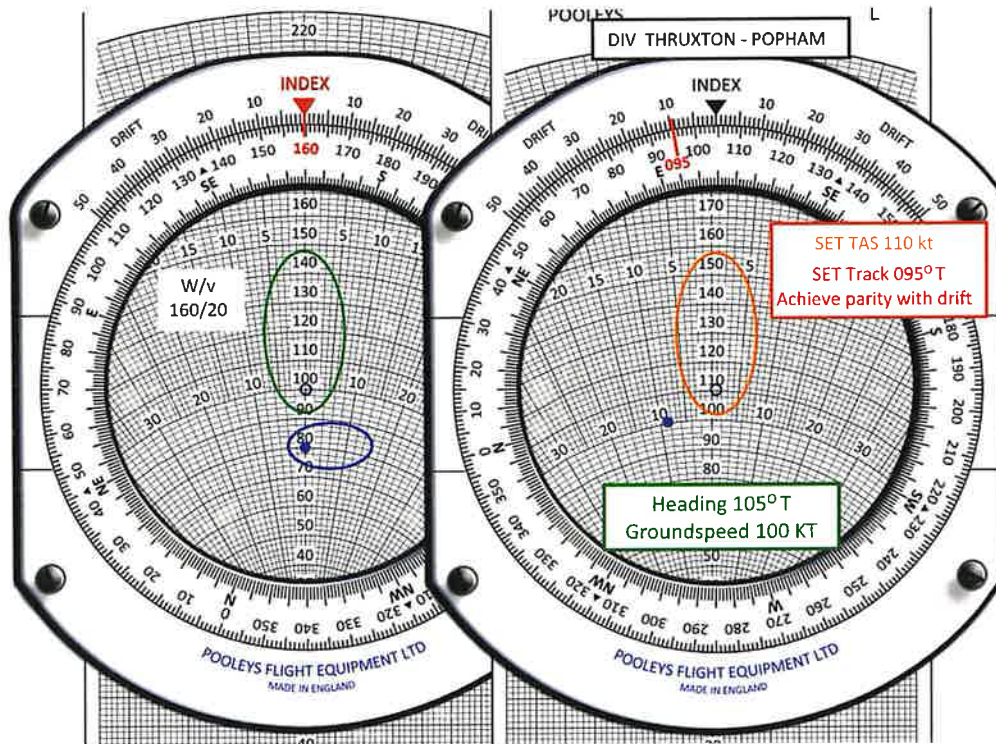
Following the same method as before we find a true heading of 201° to which we add the variation of 3°W to obtain the magnetic heading of 204° . The groundspeed is 96 knots for this leg.



The CRP can be very easy to misread, drawing a small sketch of the route will give you a rough idea of the answers you expect to obtain and ensure that any amendments are applied correctly.

**Diversion leg: Thruxton to Popham**

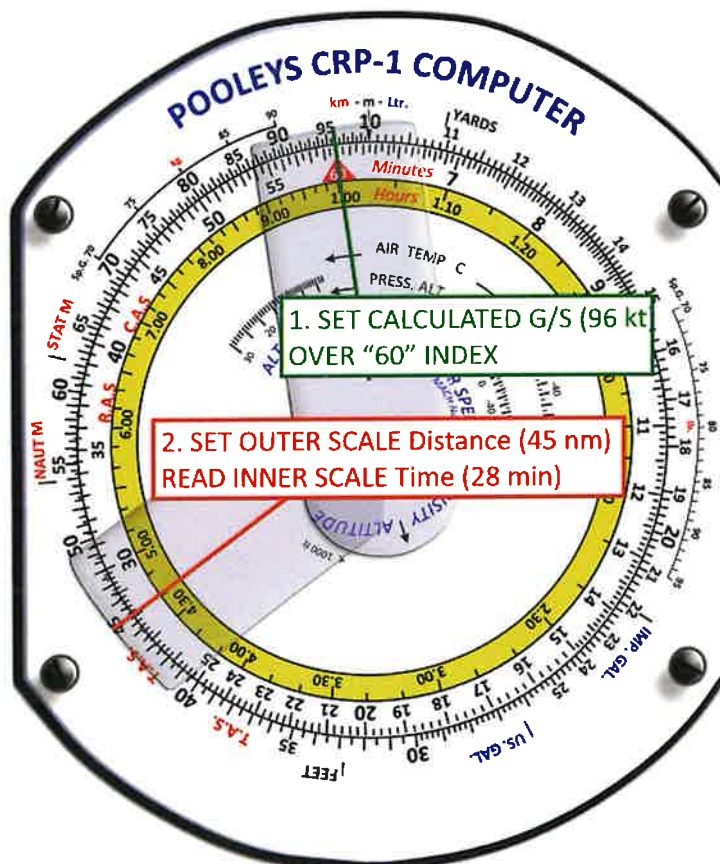
NOTE: THAT FOR THIS LEG A DIFFERENT FORECAST WIND IS USED.



To complete the plot calculations, use the front of the CRP to determine the time for each leg.

1. Set the calculated ground speed above the time index.
2. Find the distance for each leg on the outer scale.
3. Read off the time on the inner scale.

The CRP illustrated below is set for the second leg from WCO to Thruxton.



Outer Scale:	Distance	Speed
Inner Scale:	Time	60

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 QUESTIONS FOR NAVIGATION ROUTE 1

LEICESTER TO THRUXTON

1. What is the aerodrome elevation of Thruxton? Where would you find this information?
2. What type of airspace is found at 520049N 0004813W and what conditions are attached to it?
3. What is the base and class of controlled airspace (CAS) over WCO?
4. What is the meaning of the chart symbol west of WCO at Otmoor?
5. A chart symbol like the one below passes between Otmoor and Westcott. What does it denote? What action are you strongly advised to take? What station would you call?



6. What is the meaning of the group of three symbols at Westcott?
7. What type of activity takes place at Finmere, to the NNW of WCO?
8. In the vicinity of WCO what stations provide LARS?
9. Which ATSU would you plan to be in contact with at WCO?
10. When north of Thruxton which agency should you contact?
11. What is the maximum level at which to enter the Thruxton ATZ?
12. With reference to the UK AIP entry for Thruxton what is the ATS unit call sign and its hours of operation? What service does this ground station provide?
13. To the south west of Thruxton is an area denoted §127/12.0. What does this mean?
14. On the 1:500,000 chart, within each quadrangle of the graticule of latitude and longitude is a blue two digit group. What do these figures represent?
15. A fuel check over WCO reveals that you have 20 USG remaining. Assume a fuel consumption rate of 9 USG per hour. Allowing for the flight planned time to Thruxton, plus 10 minutes to the alternate Popham and for a reserve of 6 USG; what will be the safe endurance remaining on arrival at the alternate Popham?
16. To the WSW of Westcott by approximately 12 nm is Oxford airfield, it shows this symbol. What does the symbol represent and what action are you advised to take?



 END OF QUESTIONS TO ROUTE 1—LEICESTER TO THRUXTON

ANSWERS TO ROUTE 1

1. The aerodrome elevation of Thruxton is 319 ft. This can either be found on the chart, in the AIP details or the Pooley's Flight Guide.



2. R214/2.4. It is Restricted airspace from the surface to altitude 2,400 feet. It is annotated "Note 2" and looking at the chart legend it can be found that this restriction only applies to helicopters.



3. Over WCO the base of CAS is Altitude 5,500 feet, it is Class A.
4. Otmoor is a bird sanctuary. Such areas are shown with their name followed by an effective altitude. Pilots are requested to avoid these portions of airspace during certain times of the year detailed in the UK AIP.
5. This chart symbol denotes an Area of Intense Aerial Activity, in this instance the Oxford AIAA from the surface to 5,000 feet. It is strongly recommended that you contact Brize Radar 124.275 for a service, a Traffic Service would be the best option to choose.

UKAIP ENR 1.1 SECTION 5.2 HAZARDS TO FLIGHT

5.2.2 Area of Intense Air Activity (AIAA) - Airspace within which the intensity of civil and/or military flying is exceptionally high or where aircraft, either singly or in combination with others, regularly participate in unusual manoeuvres.

5.2.2.1 Intense civil and/or military air activity takes place within the areas listed in ENR 5.2. Pilots of non-participating aircraft who are unable to avoid AIAAs are to keep a good lookout and are strongly advised to make use of a radar service if available.

6. The group of symbols in the vicinity of Westcott denote a Gas Venting Operations, an NDB and a disused airfield.
7. Finmere, to the NNW of WCO is a microlight flying site.



GLIDER LAUNCH SITE
Showing maximum altitude
of winch launch AMSL



**FREE FALL PARACHUTE
DROP ZONE**



HAND/PARA GLIDING
Showing maximum altitude
of winch launch AMSL



**MICROLIGHT FLYING
SITES**

8. Brize Radar 124.275 and Farnborough LARS North 132.800.
9. Assuming they are active, you would need permission to penetrate the Benson MATZ 120.9 (or you could avoid it vertically or laterally. Check the base of CAS around Benson). As the route goes through the Oxford AIAA it is recommended that you contact Brize Radar 124.275 for a service, ideally for a Traffic Service.
10. When north of Thruxton pilots should contact either Boscombe Down 126.7 to request MATZ penetration, or Thruxton Radio ensuring that the MATZ is avoided.
From Thruxton's AIP entry:

EGHO AD 2.22 FLIGHT PROCEDURES

i. Arriving aircraft

1. Contact Boscombe Zone on 126.700 MHz for a MATZ penetration service (in accordance with ENR 2.2). Unless a direct routing is offered by Boscombe Down ATC, arriving traffic should route via a point 1 NM north of Andover. Once an aircraft has reported at this point it will be advised to free-call Thruxton A/G. Remain outside the Thruxton ATZ and contact 'Thruxton Radio' on 118.280 MHz for ATZ entry information. ATZ entry should be at 1200 FT AAL, or a maximum height to remain in VMC, not above 1200 FT AAL; or
 2. Avoid the Boscombe Down CMATZ, contact 'Thruxton Radio' for ATZ entry information. ATZ entry should be at 1200 FT AAL.
11. From the part of the Thruxton AIP entry reproduced above, we can see that the procedure promulgated for arrival has a maximum level at which to enter the ATZ of 1,200 feet. Above this Boscombe Zone on 126.700 MHz must be contacted.
 12. The call sign is Thruxton Radio operating between 0800 and 1600Z. It provides an Air/Ground Communication Service (AGCS).

AD 2.18 ATS COMMUNICATION FACILITIES

EGHO AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE number(s)	Logon Address	Hours of Operation	Remarks
1	2	3	4	5	6	7
OTHER	THRUXTON RADIO	118.280 MHz A/G frequency DOC 10 NM/3000 FT.			0900-1700 (0800-1600)	ATZ hours coincident with A/G hours.

13. The symbol denotes a Danger Area with the number 127 which is active from the surface to 12,000 feet AMSL and that there is a Danger Area Information Service is available. From the Chart legend we can find that DAIS is available from Boscombe Down.



DANGER AREA SYMBOLS



CROSSING SERVICE (DACS)



DANGER AREA INFORMATION SERVICE (DAIS)



PREFLIGHT INFORMATION AVAILABLE



BY LAWS PROHIBIT ENTRY DURING HOURS OF ACTIVITY

14. They are Maximum Elevation Figures (MEF). The MEF is shown as two figures in blue: the larger is the highest known elevation in thousands of feet above sea level; the smaller represents hundreds of feet. The figure is based on the highest known feature in each quadrangle and includes terrain, obstacles and allows for any unknown features.

The MEF provides a quick method of determining a safe altitude whilst in flight without having to study local topographical features on the chart. Generally, a safety altitude is obtained by adding a safety margin to the MEF. This is usually 1,000 ft, however the safety margin will vary depending on instructor preference or Training Organisation policy.

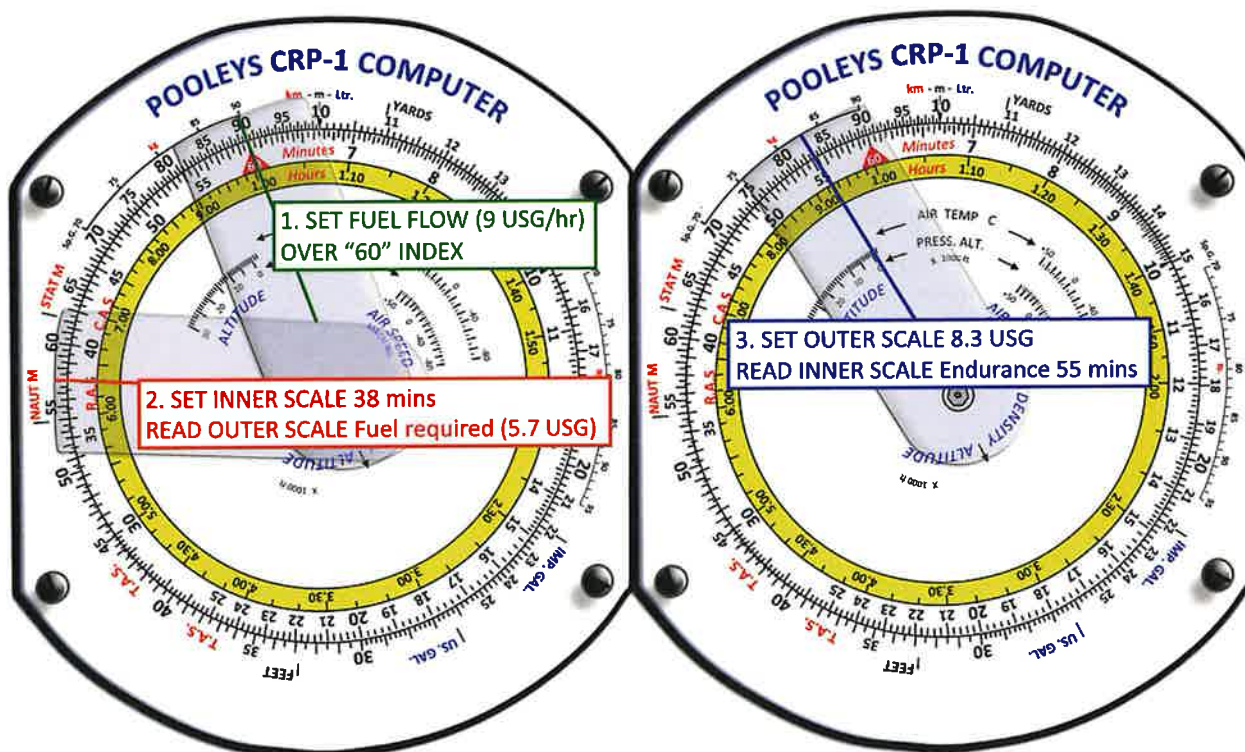
15. The safe endurance on arrival at Popham will be 55 minutes.

The reserve of 6 USG must not be taken into account, so we are left with 14 USG for the remainder of the route and flight to the diversion airfield.

From the plog Westcott to Thrupton will take 28 minutes and for the diversion we are told to allow 10 minutes; so we need to calculate how much fuel this will use and subtract it from the 14 USG we have available to use.

On the CRP set the 60 index below "90" to represent the 9 USG per hour fuel burn. A rough estimate is that our flight time is nearly 40 minutes (2/3 of an hour) and 2/3 of 9 USG is 6 USG; so we are expecting the remaining flight time to use approximately 6 USG. Read off against the 38 minute flight time on the inner scale 57 on the outer. This gives us an answer of 5.7 USG.

Subtracting this from the 14 USG available to use gives 8.3 USG. Keeping the CRP set to "90" above the index to represent the fuel flow, now read off how long 8.3 USG on the outer scale will last on the inner. Expecting roughly an hour, we see the answer on the inner scale is 55, so 55 minutes.



16. This symbol represents an instrument approach procedure outside controlled airspace, pilots are strongly advised to contact the relevant ATSU when operating within 10 nm of the aerodrome.



END OF ANSWERS TO ROUTE 1-PETERBOROUGH TO CONINGTON

NAVIGATION ROUTE 2

You are planning the following VFR flight:

From: Caernarfon (53 06 27N 004 20 42W)

To: Exeter (50 44 05N 003 24 84W)

Via: Aberystwyth (52 25 02N 004 05 00W)

and Swansea aerodrome (51 36 04N 004 04 20W)

Alternate: Dunkeswell (50 51 60N 003 14 08W)

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
CAERNAFON	ABERYSTWYTH													
53 06 27N	52 25 02N	As req.	4300	110	230/30				3.5W					
004 20 42W	004 05 00W													
ABERYSTWYTH	SWANSEA													
52 25 02N	51 36 04N	As req.	4300	110	230/30				3.5W					
004 05 00W	004 04 20W													
SWANSEA	EXETER													
51 36 04N	50 44 05N	As req.	3500	110	230/30				3W					
004 04 20W	003 24 84W													
										TOTAL				
ALTERNATE														
EXETER	DUNKESWELL													
50 44 05N	50 51 60N	As req.	2200	110	250/20				3W					
003 24 84W	003 14 08W													

Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet, within 5 nm of track rounded up to the next 100 feet

ANSWER: Completed Plog.

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
CAERNARFON	ABERYSTWYTH													
53 06 27N	52 25 02N	As req.	4900	110	230/30	168	14 P	182	3W	185	93	42.3	27.2	
004 20 42W	004 05 00W													
ABERYSTWYTH	SWANSEA													
52 25 02N	51 36 04N	As req.	4900	110	230/30	180	11 P	191	3W	194	89	49.0	33	
004 05 00W	004 04 20W													
SWANSEA	EXETER													
51 36 04N	50 44 05N	As req.	3500	110	230/30	156	15 P	171	3W	174	99	57.6	34.9	
004 04 20W	003 24 84W													
											TOTAL	148.9	95.1	
ALTERNATE														
EXETER	DUNKESWELL													
50 44 05N	50 51 60N	As req.	2200	110	250/20	043	5 S	038	3W	041	128	10.2	4.8	
003 24 84W	003 14 08W													

Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet, within 5 nm of track rounded up to the next 100 feet

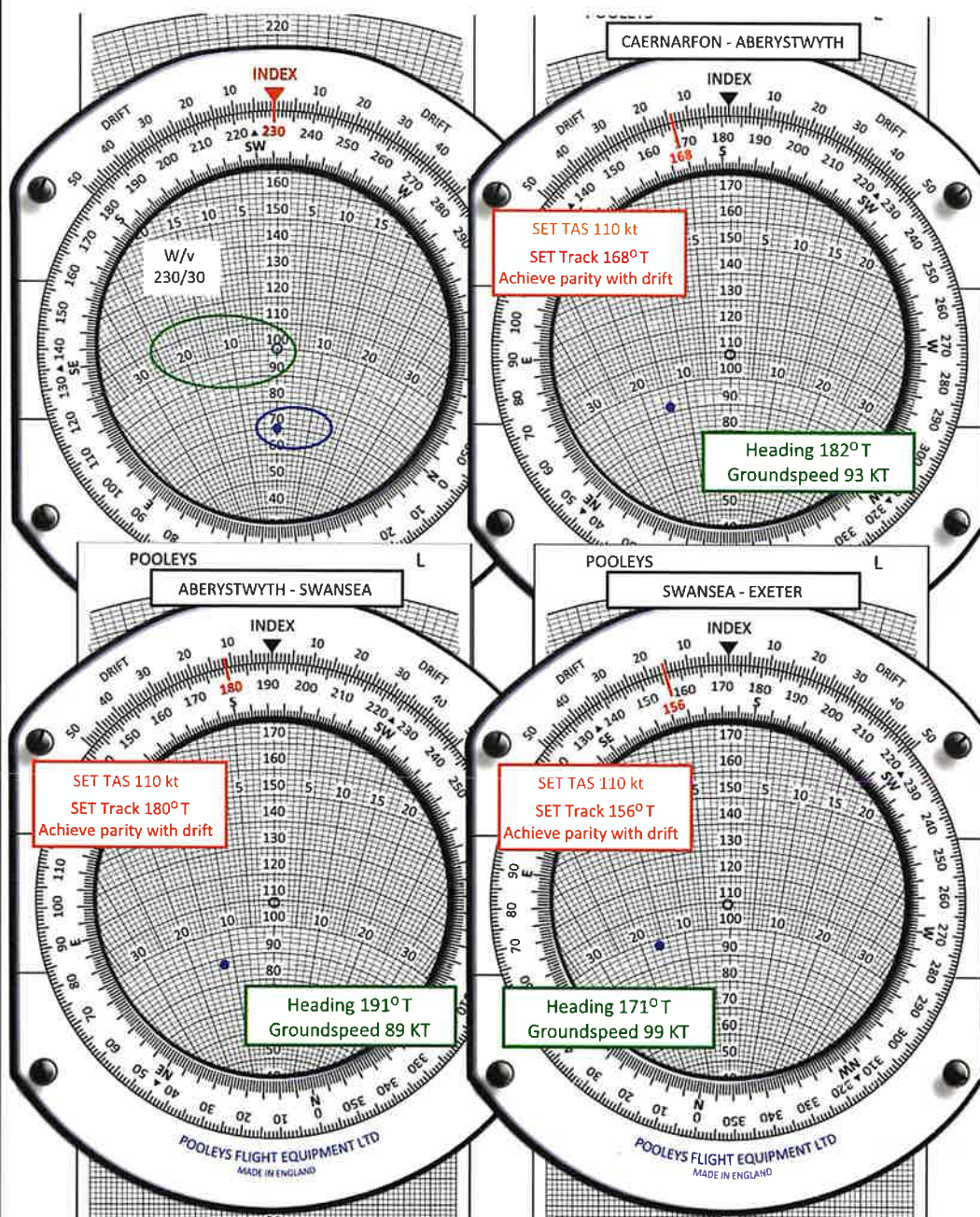
Method: Caernafon to Aberystwyth; Aberystwyth to Swansea; Swansea to Exeter

1. Place the centre of the protractor over the track drawn on the map near the mid-point between the two ends of the track.
2. Align the grid marks on the protractor with the vertical meridian lines on the chart. Read off the True Track, in this case 168°, 180° and 156°.
3. Next we need to find the effect of the wind on the aircraft. The same wind vector is forecast for all three legs, using your CRP computer set the **wind direction (230) on the rotating inner scale under the Index on the fixed outer scale.**
4. Set the centre circle on a convenient speed arc – 100 is a handy round number. Mark down the wind speed, 30 knots for this question.
5. Move the sliding scale so that the centre circle is over the TAS 110.

Leg One: Caernafon to Aberystwyth

6. Rotate the inner scale so that the measured true track, 168, is now below the index. Note the wind mark that was drawn is now displaced 15° to the left.

7. Now rotate the inner scale 15° to the left, 168 is now below 15° on the left hand side of the fixed drift scale. However, now the pencil mark indicates 14° of port drift. What we are aiming for is parity, so move the rotating scale so that 168 is below 14° on the fixed drift scale and still left of the index. In so doing the drift remains at 14° and we have parity – the track is offset 14° and the wind mark still shows 14°.
8. Read off the true heading under the Index = 182°.
9. Finally, on your plot apply the variation, in this case 3°W. Remember “East is Least, West is Best”, the variation is West, hence we add it giving a magnetic heading of 185°.
10. Read the ground speed under the wind dot – 93 knots.



Leg Two: Aberystwyth to Swansea

As the wind is the same for the second leg, rotate the inner scale so that the measured true track, 180° , is now below the index. You can repeat stages 6 to 10 above.

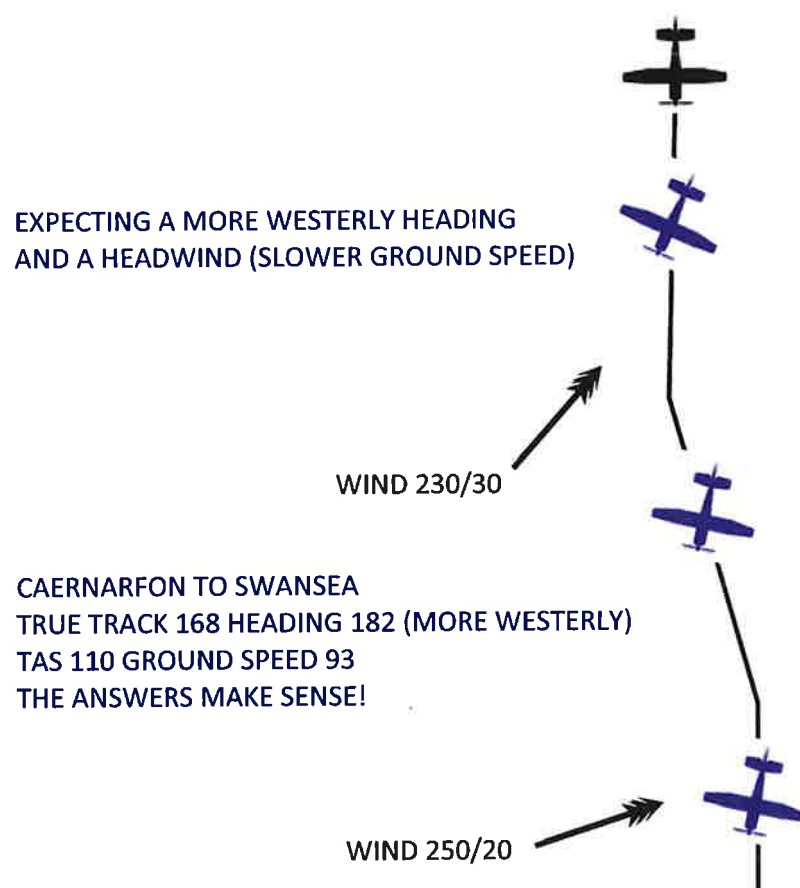
Following the same method as before we find a true heading of 191° to which we add the variation of 3°W to obtain the magnetic heading of 194° . The groundspeed is 89 knots for this leg.

Leg Three: Swansea to Exeter

As the wind is the same for the third leg, rotate the inner scale so that the measured true track, 156° , is now below the index. You can repeat stages 6 to 10 above.

Following the same method as before we find a true heading of 171° to which we add the variation of 3°W to obtain the magnetic heading of 174° . The groundspeed is 99 knots for this leg.

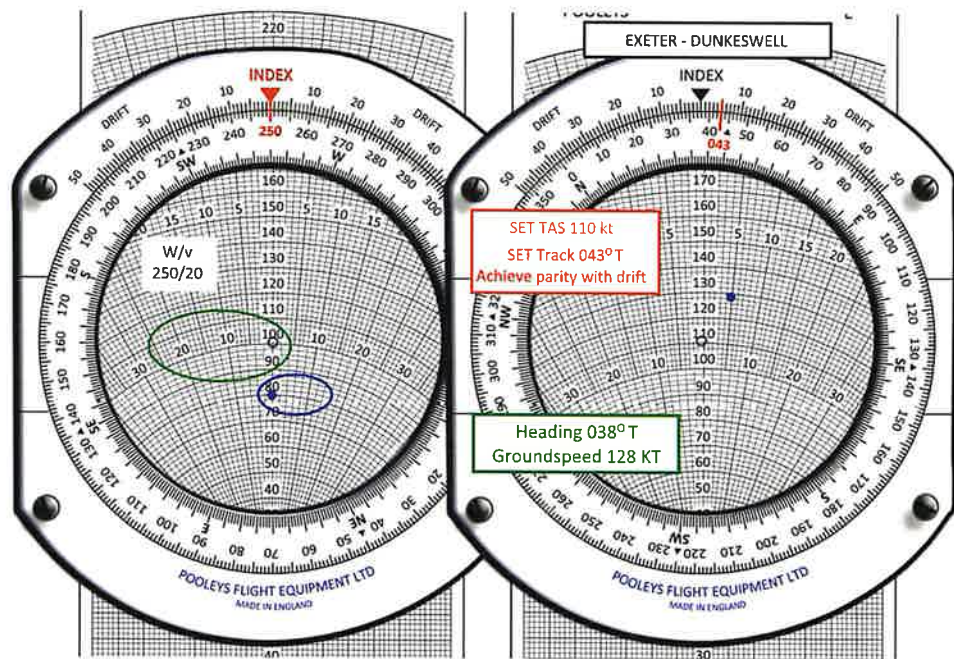
The CRP can be very easy to misread, when using it drawing a small sketch of the route is a good way to get a rough idea of the answer you expect to obtain and to ensure that any amendments are applied correctly.



Diversion: Exeter to Dunkeswell

A different wind vector is forecast for this leg and so this will need to be changed on the CRP for this leg:

1. Set the new wind direction (250) on the rotating inner scale under the Index on the fixed outer scale.
2. Set the centre circle on a convenient speed arc – 100 is a handy round number. Mark down the wind speed, 20 knots for this question.
3. Move the sliding scale so that the centre circle is over the TAS 110.



Outer Scale: Distance Speed
Inner Scale: Time 60

To complete the plog calculations, use the front of the CRP to determine the time for each leg.

1. Set the calculated ground speed above the time index.
2. Find the distance for each leg on the outer scale.
3. Read off the time on the inner scale.

 QUESTIONS FOR NAVIGATION ROUTE 2

CAERNARFON TO EXETER

1. What is the call sign of the ATS unit at Caernarfon, the type of service provided and its hours of operation?
2. Unless otherwise instructed which frequency should be contacted immediately after take-off from Caernarfon?
3. What is the meaning of the symbol at 53 04 07N 004 04 32W?
4. What service is available in relation to D217 A-K and which ATSU provides this?
5. What does the text below to the west of Aberystwyth is a marked area with the following:

***↑ D201/↑ UNL**

6. You check your fuel state overhead Swansea and you have 22 USG remaining. Assume a constant fuel consumption of 9 USG per hour and flight planned times for the subsequent legs. What will be your safe endurance on arrival at the alternate airfield, Dunkeswell? Allow a reserve of 6 USG.
7. What type of aeronautical activity takes place at Rhigos (position 51 45N 003 35W) and what is the maximum winch launch altitude?
8. What is the base of the Strumble CTA immediately above Rhigos and what class of airspace is it?
9. On a 1:500,000 chart what is the meaning of the following chart symbol?



10. What type of activity do chart symbols warn about at Dunkeswell?
11. What is the base of controlled airspace over VRP Cullompton? What Class of airspace is it?
12. What type of airspace can be found 2 nm to the SE of Exeter? What services are available and how is this information obtained?

 END OF QUESTIONS TO ROUTE 2-CAERNARFON TO EXETER

ANSWERS TO ROUTE 2

1. Caernarfon Radio, Air Ground Communication Service and 0800 to 1700.
2. EGCK AD 2.22 FLIGHT PROCEDURES Departing aircraft. Unless otherwise instructed, departing aircraft should call Valley ATC on 125.225 MHz immediately after take-off.

Service Designation	Callsign	Channel(s)	SATVOICE number(s)	Logon Address	Hours of Operation	Remarks
1	2	3	4	5	6	7
OTHER	CAERNARFON RADIO	122.255 MHz A/G frequency.			0900-1645 (0800-1545).	ATZ hours co-incident with A/G hours, but not by arrangement.

3. The highest spot elevation on each chart is marked in slightly larger print than other spot elevations and is displayed in a white rectangle with a black outline. In this instance we are looking at Mount Snowdon. In the bottom right hand corner of the chart you can find the "relief portrayal" where the highest known elevation is mentioned again.



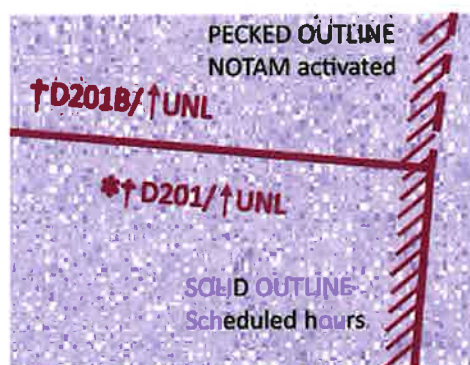
4. The multiple chart symbols denote that a DAIS "Day-iss" is available and checking the chart legend tells us that Llanbedr Tower 118.930 is the unit to call.



*D207	WEST WALES RADAR 127.090MHz
*D307	HOLBEACH RANGE 122.750MHz
	DONNA NOOK RANGE 122.750MHz
DAAIS OUTSIDE TIMES *LONDON INFORMATION 124.750MHz	
DANGER AREA ACTIVITY INFORMATION SERVICE (DAAIS) is available for certain Danger Areas (identified by the prefix S). The Nominated Air Traffic Service Units (NATSU) to be used are shown below. See UK AIP ENR 5.1. Pilots are advised to assume that a Danger Area is active if no reply is received from the appropriate NATSU.	
D001, D003, D004, D005A-B, D006A-C, D007A-C, D008A, D008C, D009A-B, *D011A-C, D012, D013, D014, *D015, D017, D021, D023 & D031	LONDON INFORMATION 124.750MHz
D026	PLYMOUTH MIL 124.150
D036	LONDON INFORMATION 124.750MHz/124.600MHz or SOUTHAMPTON or JERSEY ATSU
D037, D038, D039, D040	LONDON INFORMATION 124.750MHz/124.600MHz
*D044	LYDD INFORMATION 120.705MHz**
D061	EXETER APP 128.980MHz
D064B-C, D110, *D113A	LONDON INFORMATION 124.750MHz
*D113B, *D115A-B & *D117	
D118	PEMBREY RANGE 122.750MHz*
D119	YEOVILTON APP 127.350MHz*
D120, D122A-C	BOSCOMBE ZONE 126.700MHz
*D123, *D124, *D125	LONDON INFORMATION 124.600MHz
*D126 & *D128	
D127	BOSCOMBE ZONE 126.700MHz
D129	BRIZE ZONE 119.000MHz
*D130	FARNBOROUGH APP 125.250MHz**
*D132, D133A & D133B	FARNBOROUGH RADAR 133.440MHz**
*D136, *D138A, D138B, *D138C, *D138D	SOUTHEND APP 130.780MHz**
*D139	LONDON INFORMATION 124.600MHz
*D141	LYDD INFORMATION 120.705MHz**
D147	LONDON INFORMATION 124.750MHz
D201A-B, *D201C-D, D201F-G, *D201H-J, D202A-D	WEST WALES INFORMATION 122.155MHz
D203, D211 & D216	LONDON INFORMATION 124.750MHz
D206, D215 & D305	LONDON INFORMATION 124.600MHz
*D207	HOLBEACH RANGE 122.750MHz**
*D208	LAKENHEATH ZONE 128.900MHz
D213	COVENTRY INFORMATION 123.830MHz**
D217A-K	LLANBEDR TWR 118.930MHz
*D304 & D314	MANCHESTER APP 118.580MHz
*D307	DONNA NOOK RANGE 122.750MHz**
LEGEND NOTES	
OUTSIDE TIMES { * LONDON INFORMATION 124.750MHz or ** LONDON INFORMATION 124.600MHz	
1. DANGER AREA D129/FL120 WESTON ON THE GREEN. Is extensively used for Military Parachute Training. Aircraft can be encountered on 'Run-In' routes outside the Danger Area boundary, particularly up to 2500ft ALT. Pilots are strongly advised to maintain constant vigilance when transiting adjacent airspace. DAAIS available on BRIZE ZONE 119.000MHz.	
2. RESTRICTED AREAS. Restricted Areas R107, R204, R212 and R214 apply only to Helicopters.	
3. AREAS OF INTENSE AERIAL ACTIVITY. Night operations may be conducted by aircraft using reduced navigation and/or anti-collision lights, at the following areas: Shawbury, Yeovilton, Culdrose and Salisbury Plain.	
4. MILITARY HELICOPTER OPERATIONS IN THE SALISBURY PLAIN AREA. In addition to intensive daytime activities, helicopters may be encountered operating during the hours of darkness without, or with restricted navigation lights. For vertical and lateral limits see UK AIP ENR 1.1.	
5. KEEVIL AERODROME (511850N 0020643W). In addition to use as a Glider Launching Site, Keevil is used extensively as a Military Dropping Zone. Pilots are advised to avoid the aerodrome at all times by 2NM laterally and 2000ft AAL vertically. (Keevil elevation is 200ft AMSL).	
6. HOLYHEAD CTA. Provision of ATS within these sectors is deferred to the relevant NOTAM.	

5. D201 is a Danger Area number 201 with a Danger Area Crossing Service (DACS) available, it is active during scheduled hours and has no upper limit.

The solid outline denotes a Danger Area that is active during scheduled hours, areas activated by NOTAM have a pecked outline. After the oblique the level is stated, in this instance UNL meaning unlimited, i.e. there is no upper level.



DANGER AREA SYMBOLS

† CROSSING SERVICE (DACS)

S DANGER AREA INFORMATION SERVICE (DAAIS)

T PREFLIGHT INFORMATION AVAILABLE

* BY LAWS PROHIBIT ENTRY DURING HOURS OF ACTIVITY

6. The flight planned times for the subsequent legs are:

Swansea to Exeter: 34.9 minutes

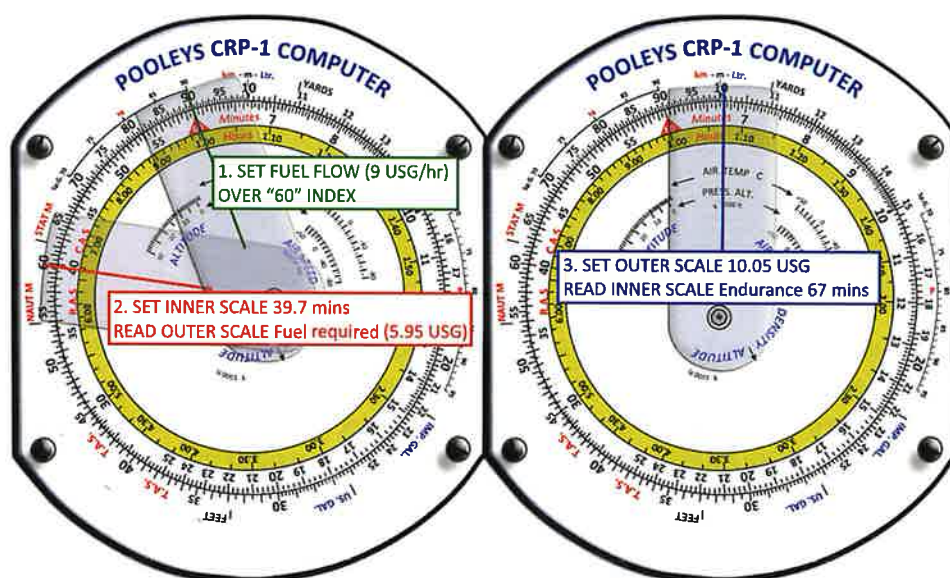
Exeter to Dunkeswell: 4.8 minutes

Total: 39.7 minutes

Firstly, find out how much fuel the remaining flight time requires, using the CRP set the 60 index below 9 for the fuel flow. Read off above the time (39.7 minutes) on the inner scale a fuel requirement of 5.95 USG. A rough calculation will confirm whether we have the decimal point in the correct place: 40 minutes is $\frac{2}{3}$ of an hour, $\frac{2}{3}$ of 9 USG is 6.

Overhead Swansea we had 22 USG of fuel and from this we need to subtract the 5.95 USG we are going to use and the 6 USG that we are told in the question must be held in reserve:
 $22 - 5.95 - 6 = 10.05$ USG

Keeping the CRP as above with the 60 index below the 9, this time read off below 10.05 on the outer scale a time of 67 minutes on the inner scale. Therefore the endurance remaining is 1 hour and 7 minutes.



7. Gliding takes place at Rhigos with winch launching up to 2,800 feet amsl.

In relation to gliding sites, the vertical extent of the activity shown on the chart is the maximum altitude (i.e. amsl) that the gliders launch to. Below that level winch wires or tugs (with or without a glider in tow!) can be expected. Gliding activity should be expected well above the level indicated (this is merely the glider launch level), and in the vicinity as glider flying is not confined to the circle drawn on the chart.



GLIDER LAUNCH SITE

Showing maximum altitude of winch launch AMSL



FREE FALL PARACHUTE DROP ZONE



HAND/PARA GLIDING

Showing maximum altitude of winch launch AMSL



MICROLIGHT FLYING SITES

8. The Strumble CTA is Class A controlled airspace extending from FL 75. The annotation FL 75(+) means that the airspace extends to an upper limit of FL 195.
9. This symbol denotes the boundary between Altimeter Setting Regions.
10. Dunkeswell has free fall parachute drop zone within its ATZ (the solid circle). Such areas may be either 1.5 nm or 2 nm in radius, within them parachutists should be expected up to FL150.
11. Above the Cullompton VRP is the Berry Head CTA, the base is FL65 and it is Class A airspace. Note that here a minimum altitude is specified. Giving a minimum altitude for the base is designed to help prevent airspace infringements. For example with a QNH of 970 hPa FL65 would be equivalent to 5210 ft, but the chart tells us that no lower than altitude 5,500 feet will be used.
12. 2 nm SE of Exeter is a danger area extending from the surface to altitude 1,500 feet.
From the chart we can see that a DAIS is available, as well as pre-flight information. The asterisk shape also tells us that entry is prohibited by law during the hours of activity, making use of the first two services is therefore a good idea!

The chart legend tell us that Exeter Approach 128.980 are the ATSU to contact for DAIS or there is a telephone number listed for pre-flight information. You can also see that the name and elevation of Exeter airport are surrounded by a pecked line, this tells us that Exeter is a Customs Aerodrome.



END OF ANSWERS TO ROUTE 2-CAERNARFON TO EXETER

NAVIGATION ROUTE 3

You are planning the following VFR flight:

From: Bodmin (50 30 06N 004 39 96W)

To: Wycombe (51 36 70N 000 48 49W)

Via: Melksham (51 22 45N 002 08 30W)

Alternate: Blackbushe (51 19 42N 000 50 90W)

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
BODMIN	MELKSHAM													
50 30 06N	51 22 45N	As req.	2800	100	330/35				3W					
004 39 96W	002 08 30W													
MELKSHAM	WYCOMBE													
51 22 45N	51 36 70N	As req.	3300	100	320/25				2W					
002 08 30W	000 48 49 W													
ALTERNATE										TOTAL				
WYCOMBE	BLACKBUSHE													
51 36 70N	51 19 42N	As req.	1300	100	320/25				2W					
000 48 49W	000 50 90W													

Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet, within 5 nm of track rounded up to the next 100 feet

ANSWER: Completed Plog.

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
BODMIN	MELKSHAM													
50 30 06N	51 22 45N	As req.	2800	100	330/35	061	20 S	041	3W	044	95	109	69	
004 39 96W	002 08 30W													
MELKSHAM	WYCOMBE													
51 22 45N	51 36 70N	As req.	3300	100	320/25	074	13 S	061	2W	063	107	51.7	29	
002 08 30W	000 48 49 W													
										TOTAL		160.7	98	
ALTERNATE														
WYCOMBE	BLACKBUSHE													
51 36 70N	51 19 42N	As req.	1300	100	320/25	185	10 P	195	2W	197	116	17.3	9.1	
000 48 49W	000 50 90W													

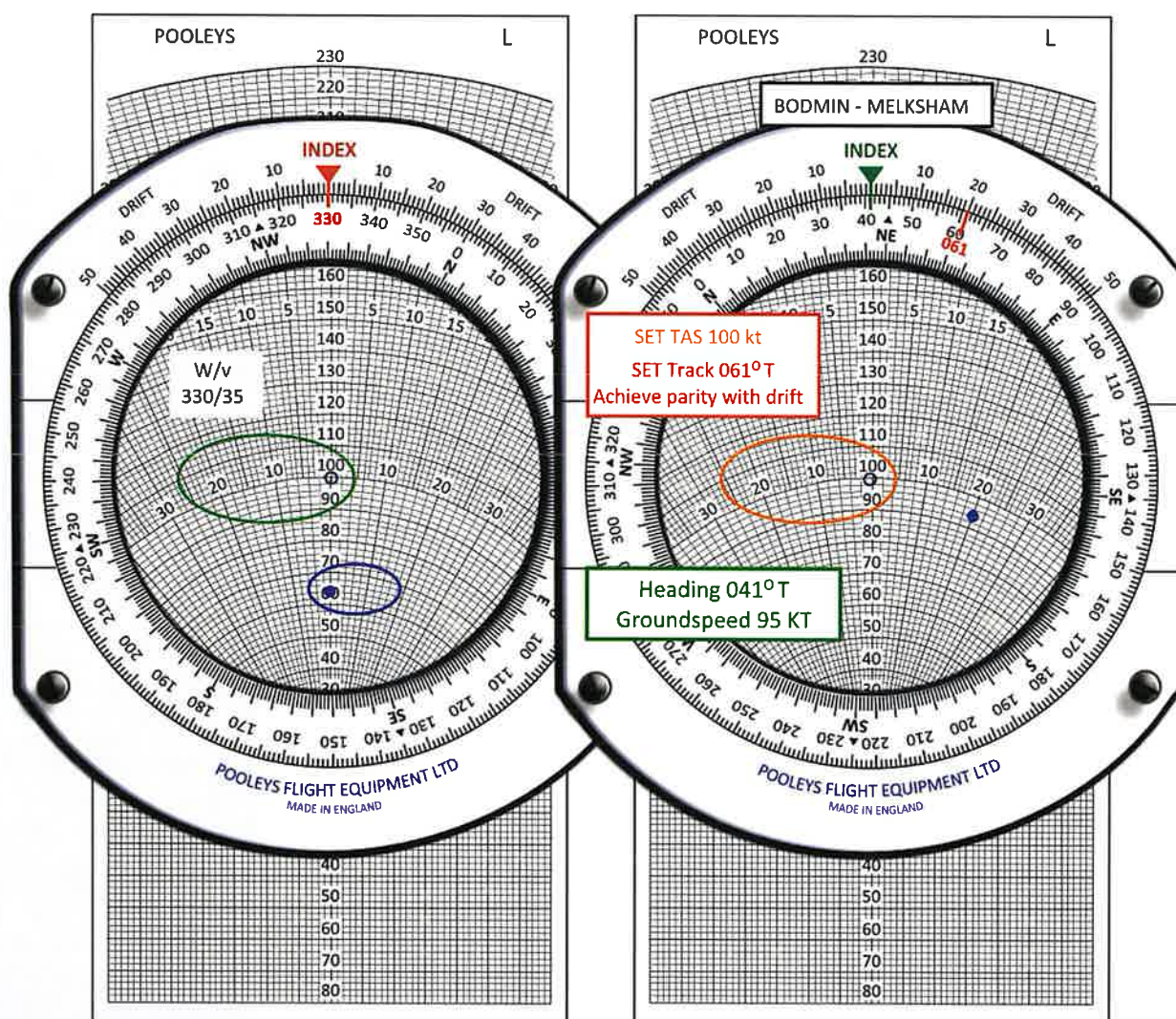
Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet; within 5 nm of track rounded up to the next 100 feet

METHOD:

1. Place the centre of the protractor over the track drawn on the map near the mid-point between the two ends of the track.
2. Align the grid marks on the protractor with the graticule lines on the chart. Read off the True Track, in this case 061°, 074° and 185°.

FIRST LEG: Bodmin to Melksham;

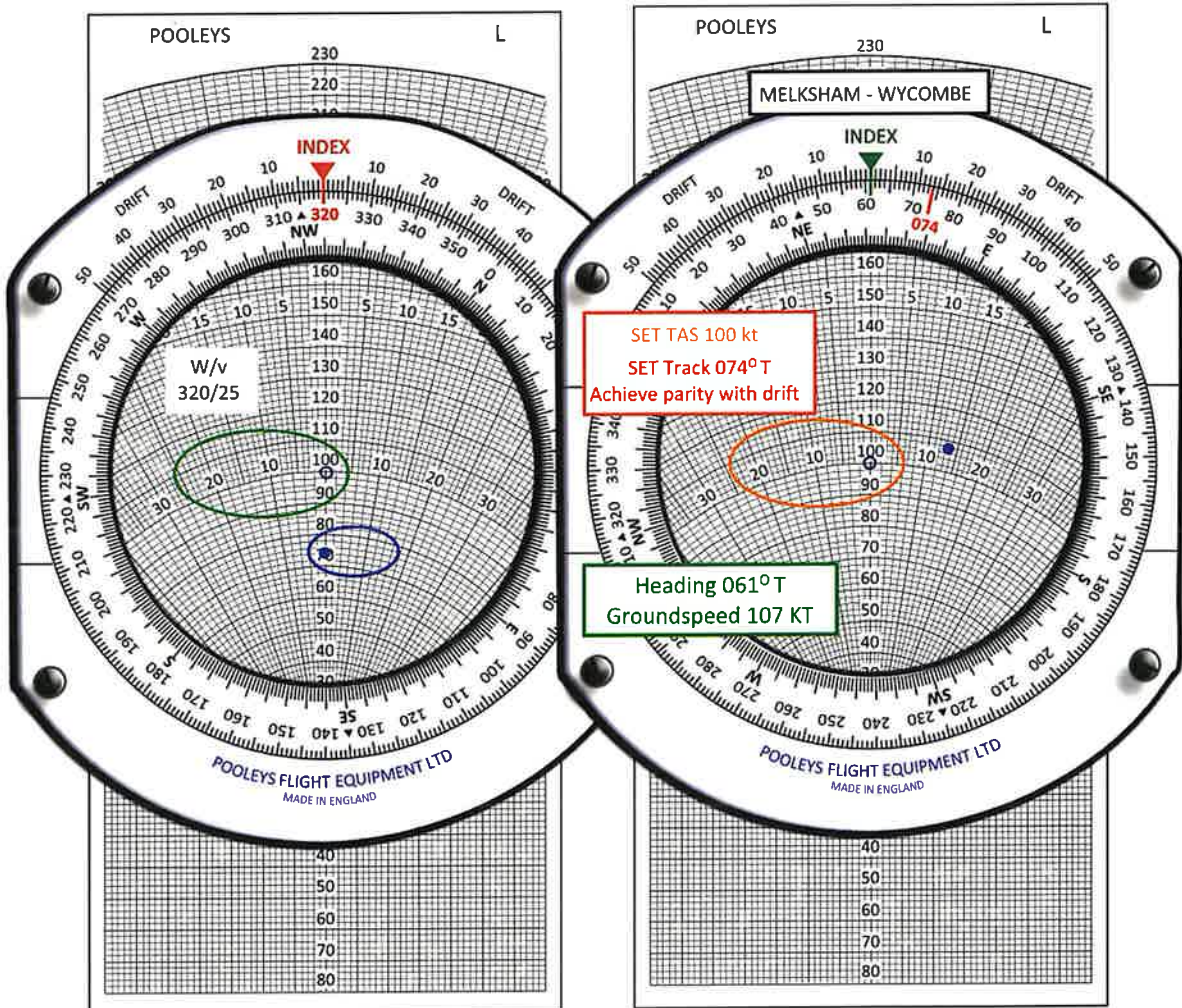
3. Next, find the effect of the wind on the aircraft. Using the CRP computer set the **wind direction (330°)** on the **rotating inner scale** under the **Index** on the **fixed outer scale**.
4. Set the **centre circle** on a convenient speed arc – **100** is a handy round number. Mark down the wind speed, **35** knots for this route.
5. The **TAS** is **100**, so the **centre circle** is already in the correct place.
6. Rotate the inner scale so that the measured true track, **061°**, is now below the index. Note the wind mark is now displaced **20°** to the right
7. Rotate the inner scale **20°** to the right, **061°** is now below **20°** on the right hand side of the fixed drift scale. We are aiming for parity, so ensure that the wind dot is still displaced **20°**.
8. Read off the true heading under the Index = **041°**
9. Finally apply the variation, in this case **3°W**. Remember "East is Least, West is Best", the variation is West, hence we add it giving a magnetic heading of **044°**
10. Read off the groundspeed below the wind dot – **95** knots.



SECOND LEG: Melksham to Wycombe

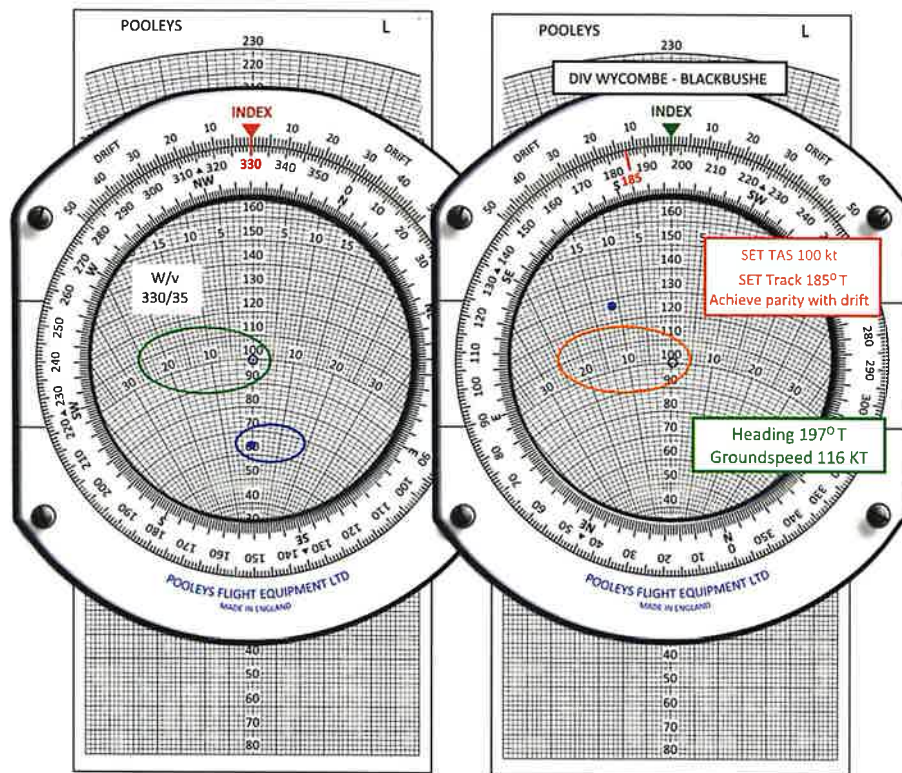
NOTE: A DIFFERENT FORECAST WIND IS USED AND WE MUST "RESET" THE CRP

3. Set the wind direction (320) on the rotating inner scale under the Index on the fixed outer scale.
4. Set the centre circle on a convenient speed arc – 100 is a handy round number. Mark down the wind speed, 25 knots for this route.
5. The TAS is 100, so the centre circle is already in the correct place.
6. Rotate the inner scale so that the measured true track, 074° , is now below the index. Note the wind mark is now displaced 14° to the right.
7. Rotate the inner scale 14° to the right, 074° is now below 14° on the right hand side of the fixed drift scale. We are aiming for parity, but the wind dot now indicates 13° , so move 074° under 13° on the fixed drift scale. Check that the wind dot is still displaced 13° .
8. Read off the true heading under the Index = 061° .
9. Finally apply the variation, in this case 2° W. Remember "East is Least, West is Best", the variation is West, hence we add it giving a magnetic heading of 063° .
10. Read off the groundspeed below the wind dot – 107 knots.

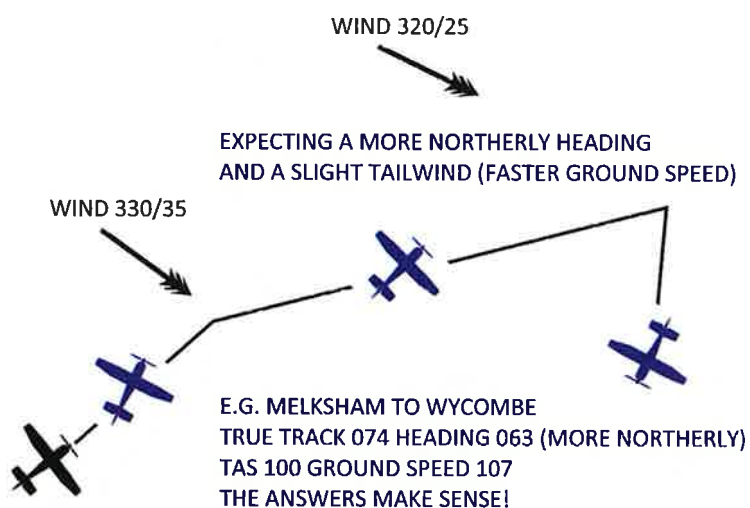


Diversion: Wycombe to Blackbushe

The diversion uses the same wind as leg two, so set the measured track of 185° below the index and repeat stages 6 to 10. This will give a magnetic heading of 197° and a ground speed of 116 knots.



The CRP can be very easy to misread, making a small sketch will give you a rough idea of the answer you expect to obtain to ensure that any amendments are applied correctly.



To complete the plog calculations, use the front of the CRP to determine the time for each leg.

1. Set the calculated ground speed above the time index.
2. Find the distance for each leg on the outer scale.
3. Read off the time on the inner scale.

Outer Scale:	Distance	Speed
Inner Scale:	Time	60

QUESTIONS FOR NAVIGATION ROUTE 3

BODMIN TO WYCOMBE

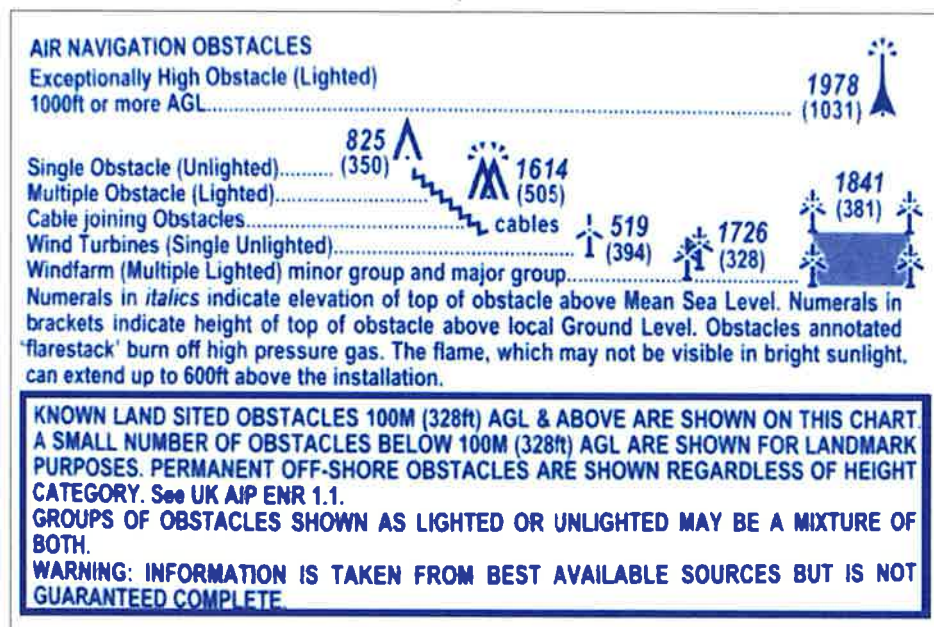
1. What ASR is Bodmin in?
2. 9 nm east of Bodmin is the following symbol: What does it mean?
3. What feature can be seen at 51 01.11N 003 06.03W?
4. What activity takes place at 50 21.90N 003 42.52W?
5. What does the symbol representing Benson Aerodrome (51 36.85N 001 05.85W) mean?
6. Looking to the west of Melksham, which unit would you call if you wished to enter the controlled airspace?
7. What is the base of CAS directly over Wycombe airfield?
8. What is the airfield elevation at Blackbushe?
9. On the diversion leg from Wycombe to Blackbushe, what is the lowest base of controlled airspace? What type of airspace is it?
10. Consider the entire route and diversion leg. Which agencies are available to provide radar services to aircraft outside controlled airspace?



END OF QUESTIONS TO ROUTE 3-BODMIN TO WYCOMBE

ANSWERS TO ROUTE 3

1. Bodmin is in the Wessex ASR.
2. The symbol represents multiple lit obstacles.



3. The feature at 51 01.11N 003 06.03W is a large town or city, in this case Taunton.

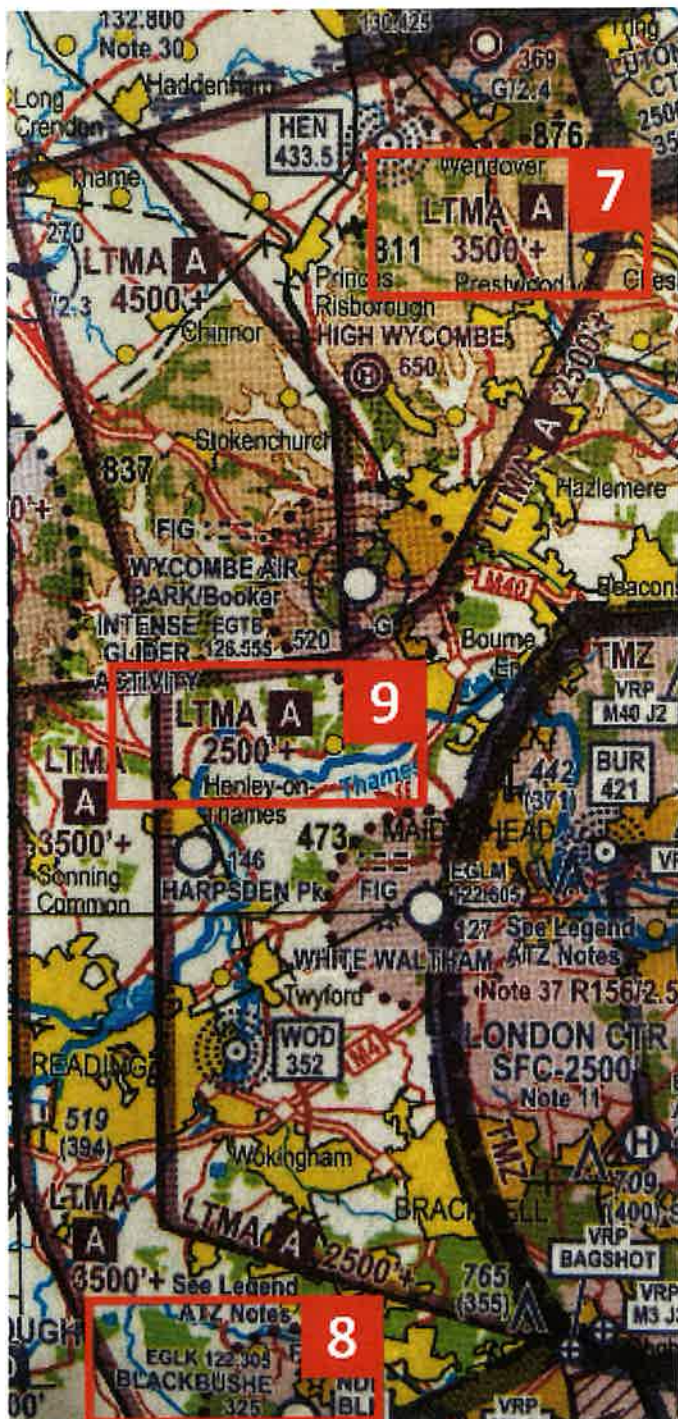
BUILT-UP AREAS

- City or large Town  over 1 Sq km
- Small Town, Village or Hamlet  under 1 Sq km
- Large Industrial Area 

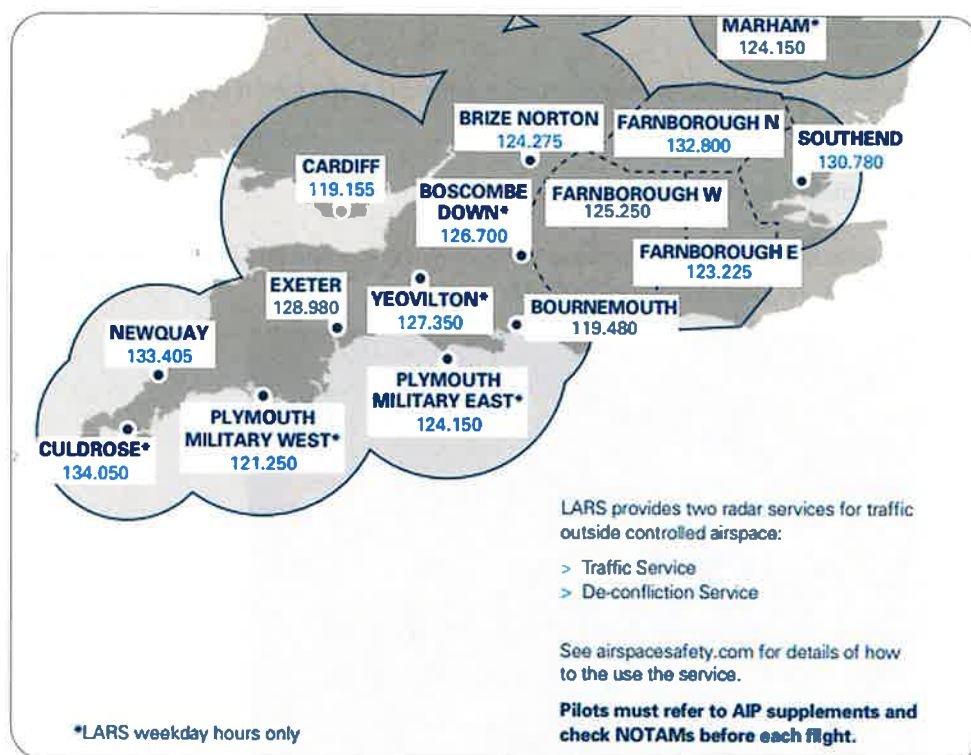
4. Microlight flying takes place at Halwell, 50 21.90N 003 42.52W, see below.

-  AERODROME - Civil
-  AERODROME - Civil, limited or no facilities
-  HELIPORT - Civil
-  AERODROME - Government, available for Civil use
-  AERODROME - Government
-  HELIPORT - Government
-  MICROLIGHT FLYING SITES
-  DISUSED or ABANDONED Aerodrome
-  AERODROME - Training Aerodrome: flight training, including circuit training, takes place from this aerodrome.
-  AERODROME - Unusual Activity: unusual aerial activities, including Aerobatic and formation flights, take place from this aerodrome.

5. The symbol at Benson (51 36.85N 001 05.85W) denotes a Government Aerodrome available for Civil use.
6. To obtain permission to transit the Bristol Control Zone you should contact Bristol Radar on 125.650 MHz.
7. The base of CAS directly over Wycombe airfield is 3,500 feet, it is Class A and part of the London TMA.
8. The airfield elevation at Blackbushe is 325 feet.
9. The lowest base of controlled airspace between Wycombe and Blackbushe is 2,500 feet, it is Class A and part of the London TMA.



10. The LARS units along the route are: Newquay, Exeter, Yeovilton, Brize Norton and Farnborough LARS West (you could also include Plymouth Military East and West and Farnborough LARS North, although these are less appropriate for the route).



END OF ANSWERS TO ROUTE 3 – BODMIN TO WYCOMBE

NAVIGATION ROUTE 4

You are planning the following VFR flight:

From: Rochester (51 21 07N 000 30 30E)

To: Peterborough/Conington (52 27 90N 000 15 28W)

Via: Elmsett (52 04 51N 000 58 65E)

Alternate: Bourn (52 12 88N 000 02 47W)

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
ROCHESTER	ELMSETT													
51 21 17N	52 04 51N	As req.	1900	95	040/20				2W					
000 30 30E	000 58 65E													
ELMSETT	CONINGTON													
52 04 51N	52 27 90N	As req.	1600	95	040/20				2W					
000 58 65E	000 15 28W													
										TOTAL				
ALTERNATE														
CONINGTON	BOURN													
52 27 90N	52 12 88N	As req.	1300	95	050/15				2W					
000 15 28W	000 02 47W													

Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet, within 5 nm of track rounded up to the next 100 feet

ANSWER: Completed Plog.

TAKE OFF:

LAND:

FLIGHT TIME:

FROM	TO	ALT/FL	SAFE ALT	TAS	W/V	TRK T	DRIFT	HDG T	VAR	HDG M	GS	DIST	TIME	ETA
ROCHESTER	ELMSETT													
51 21 07N	52 04 51N	As req.	1900	95	040/20	023	4 P	026	2W	028	75	46.8	37.4	
000 30 10E	000 58 65E													
ELMSETT	CONINGTON													
52 04 51N	52 27 90N	As req.	1600	95	040/20	298	12 P	310	2W	312	97	50.9	31.5	
000 58 65E	000 15 28W													
											TOTAL	97.7	68.9	
ALTERNATE														
CONINGTON	BOURN													
52 27 90N	52 12 88N	As req.	1300	95	050/15	154	9 S	145	2W	147	98	16.9	10.4	
000 15 28W	000 02 47W													

Note: Safety Altitude is calculated from the higher of: i) the highest ground plus 1,299 feet; or ii) the highest structure plus 1,000 feet, within 5 nm of track rounded up to the next 100 feet

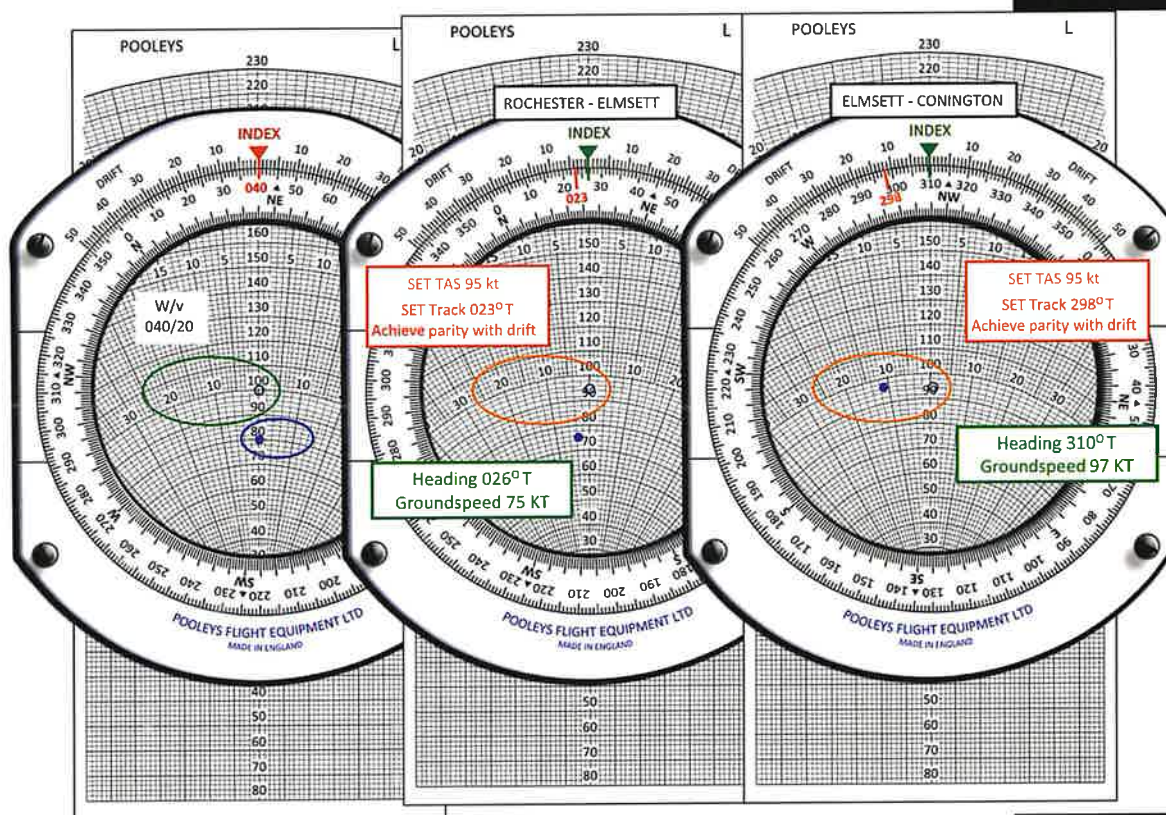
Method: Rochester to Elmsett; Elmsett to Conington

1. Place the centre of the protractor over the track drawn on the map near the mid-point between the two ends of the track.
2. Align the grid marks on the protractor with the graticule lines on the chart. Read off the True Track, in this case 023°, 298° and 154°.
3. Next, find the effect of the wind on the aircraft. Using the CRP computer set the **wind direction for the first two legs (040) on the rotating inner scale under the Index on the fixed outer scale.**
4. **Set the centre circle on a convenient speed arc – 100 is a handy round number.** Mark down the wind speed, 20 knots for this question.

Leg One: Rochester to Elmsett

5. **Move the sliding scale so that the centre circle is over the TAS 95.**
6. Rotate the inner scale so that the measured true track, 023°, is now below the index. Note the wind mark that was drawn is now displaced 5° to the left.

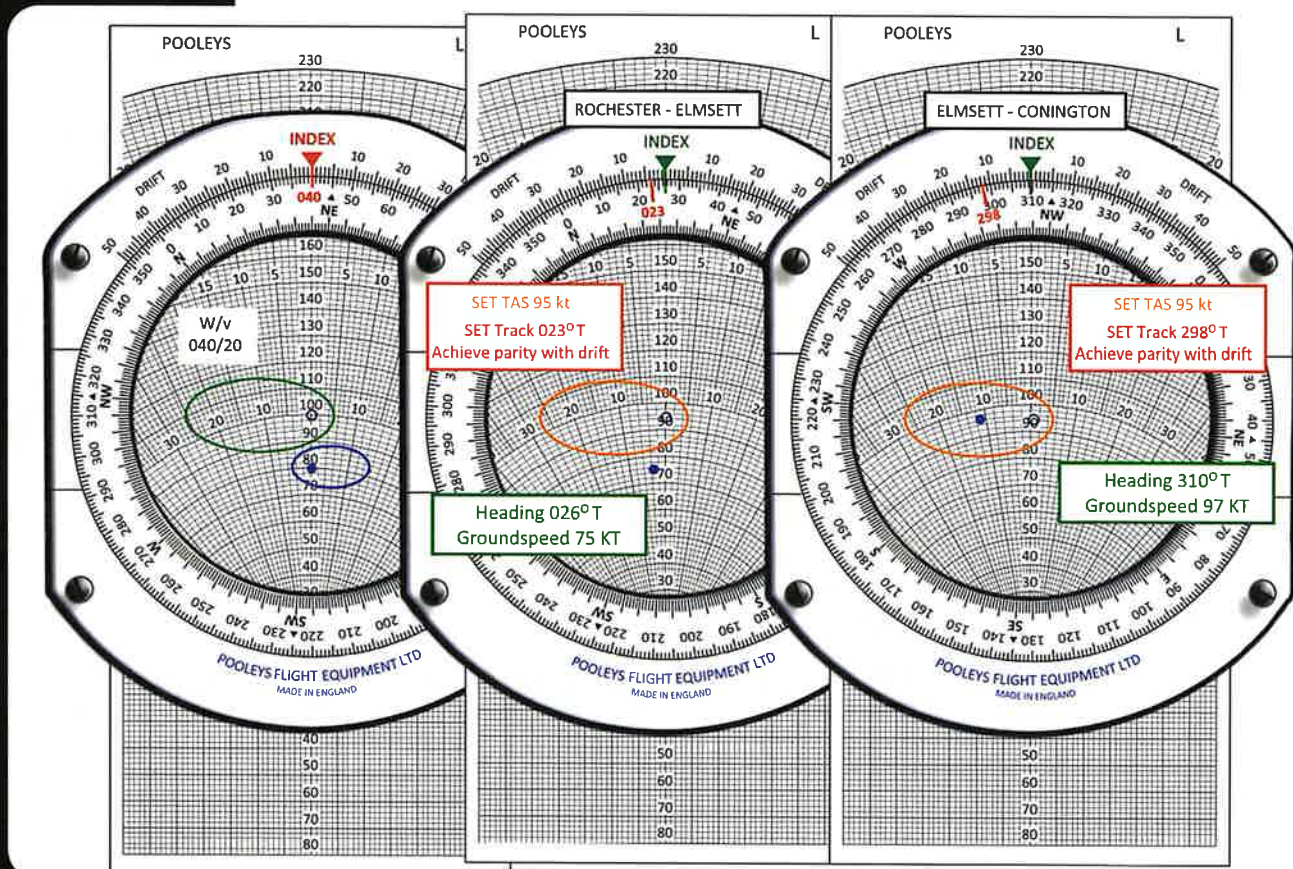
7. Now rotate the inner scale 5° to the left, 023° is now below 5° on the left hand side of the fixed drift scale. However, the wind dot is now only displaced by 4°; adjust the inner scale so that 023° is now under 4° on the fixed drift scale. Check to ensure that the wind dot is still displaced 4° to the left. We are aiming for parity, so if 023° is below 4° on the fixed drift scale and the wind dot is also displaced by 4° then the computer is set correctly.
8. Read off the true heading under the Index = 026°.
9. Finally apply the variation, in this case 2°W. Remember "East is Least, West is Best", the variation is West, hence we add it giving a magnetic heading of 028°.
10. Read the groundspeed below the wind dot – 75 knots.



Leg Two: Elmsett to Conington

The wind vector for this leg is the same and so we can use the wind as set for the first leg.

1. Rotate the inner scale so that the measured true track, 298°, is now below the index.
2. Note the wind mark that was drawn is now displaced 12° to the left.
3. Rotate the inner scale 12° to the left, 298° is now below 12° on the left hand side of the fixed drift scale. The wind dot is still indicating 12° of port drift, so we have parity between the wind dot and the displacement of the true track.
4. Read off the true heading under the Index = 310°.
5. Finally apply the variation, in this case 2°W. Remember "East is Least, West is Best", the variation is West, hence we add it giving a magnetic heading of 312°.
6. Read the groundspeed below the wind dot – 97 knots.



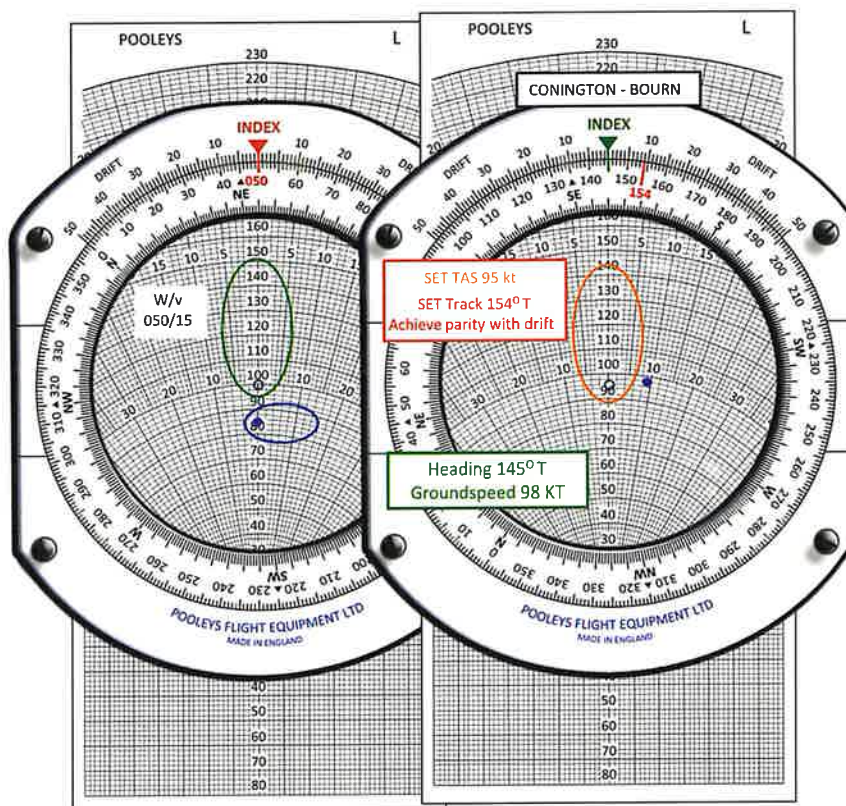
Diversion: Conington to Bourn

For this leg a new wind vector must be drawn.

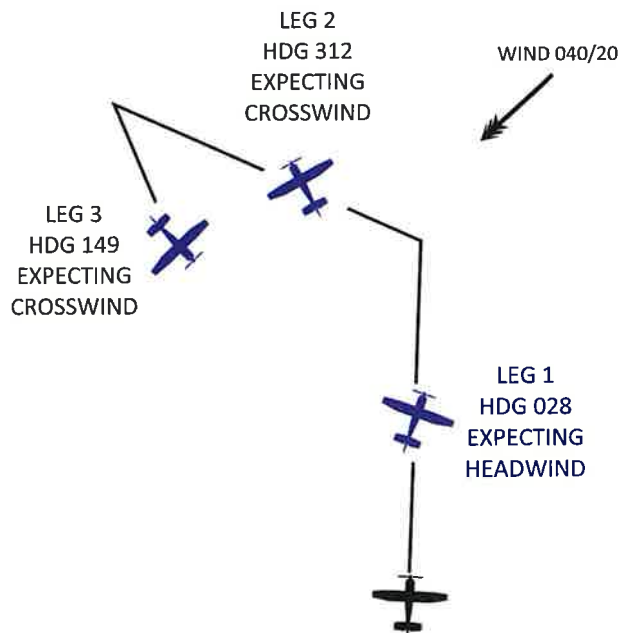
1. To find the effect of the wind on the aircraft. Using the CRP computer set the **wind direction (050°)** on the rotating inner scale under the **Index** on the fixed outer scale.
2. Set the centre circle on a convenient speed arc – 100 is a handy round number. Mark down the wind speed, 15 knots for this question.
3. Move the sliding scale so that the centre circle is over the TAS 95.
4. Rotate the inner scale so that the measured true track, 154°, is below the index.

NOTE: THE WIND MARK THAT WAS DRAWN IS NOW DISPLACED 9° TO THE RIGHT.

5. Rotate the inner scale 9° to the right, 154° is now below 9° on the right hand side of the fixed drift scale. The wind dot is still indicating 9° of starboard drift, so we have parity between the wind dot and the displacement of the true track.
6. Read off the true heading under the Index = 145°.
7. Finally apply the variation, in this case 2°W. Remember "East is Least, West is Best", the variation is West, hence we add it giving a magnetic heading of 147°.
8. Read the groundspeed below the wind dot – 98 knots.



The CRP can be very easy to misread, drawing a small sketch will give you a rough idea of the answer you expect to obtain to ensure that any amendments are applied correctly.



To complete the plog calculations, use the front of the CRP to determine the time for each leg.

1. Set the calculated ground speed above the time index.
2. Find the distance for each leg on the outer scale.
3. Read off the time on the inner scale.

END OF PRACTICE PLANNING AND PLOGS

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QUESTIONS FOR NAVIGATION ROUTE 4

PETERBOROUGH TO CONINGTON

1. In the vicinity of Rochester what agencies are able to provide LARS services?
2. To the west of the Sheerness VRP are two overlapping magenta circles. What do they indicate?
3. The turning point at Elmsett lies within a light blue shaded area surrounded by darker blue dots. What does this area denote and which agency should you contact and why?
4. What types of airspace is established around Stansted?
5. TMZ question.
6. Next to several of the airspace descriptions around Stansted there is a four digit number in brackets. What does this represent?
7. What is the meaning of the following symbol? 
8. Along the second leg from Elmsett, you call Lakenheath to request permission to transit the MATZ. No reply is received, what is the best course of action to take?
9. What type of airfield is Moleseworth?
10. What types of airspace is shown by a circle, 1 nm radius, centred at 520621N 0002521W?

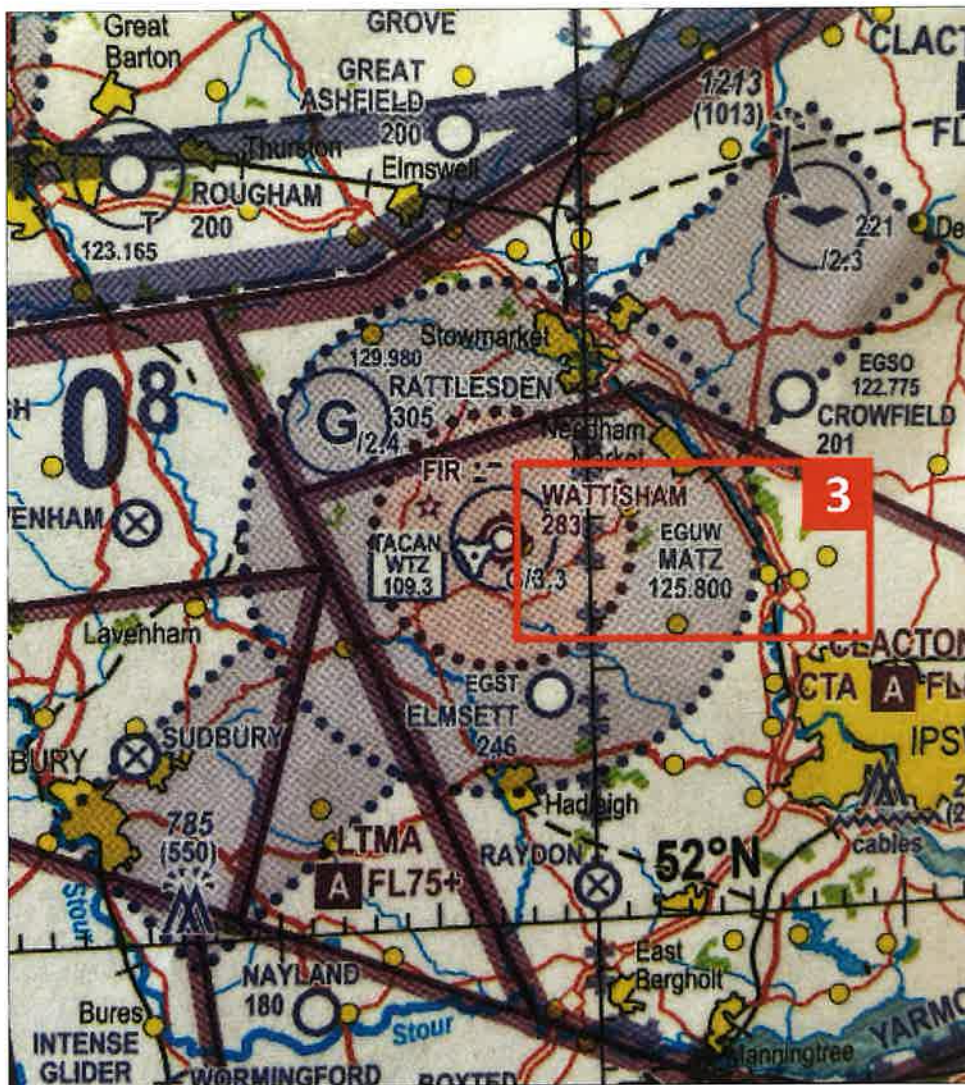
END OF QUESTIONS TO ROUTE 4-PETERBOROUGH TO CONINGTON

ANSWERS TO ROUTE 4

1. The designated LARS units in the vicinity of Rochester are Southend 130.780 MHz Farnborough LARS North 132.800 and, Farnborough LARS East 123.225 MHz.
2. They are Gas Venting Stations up to altitude 3,500 feet. These stations can create severe turbulence during the venting of natural gas under high pressure. Pilots are advised to avoid flying below the maximum vertical limit and within the boundary of the areas shown on the chart.



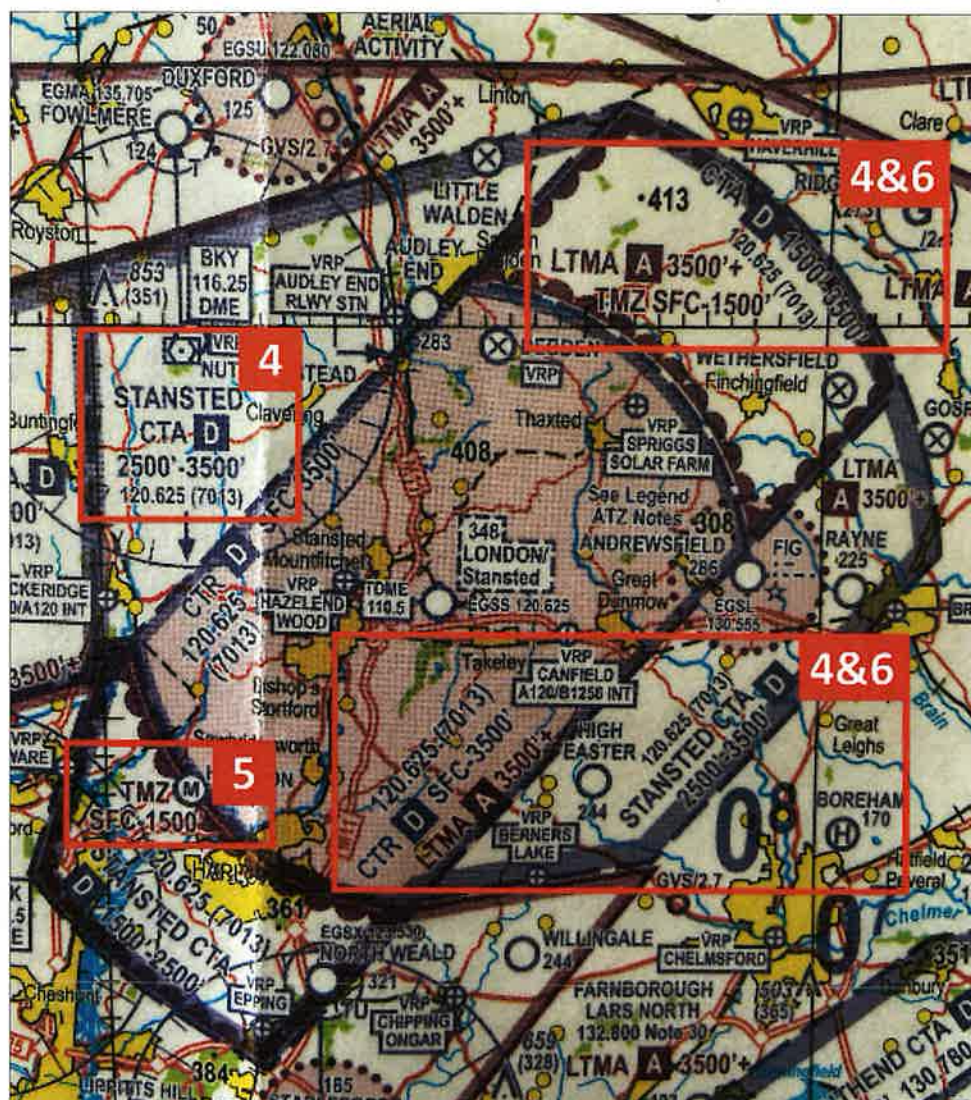
3. The area is a MATZ (Military Aerodrome Traffic Zone); pilots should contact Wattisham 125.8 MHz when 15 nm or 5 minutes flying time from the MATZ boundary (whichever is sooner) to request permission to penetrate the MATZ.



4. Stanstead has a Class D Control Zone (CTR) from the surface to altitude 3,500 feet, a Class D Control Area (CTA) to the west between 2,500 feet and 3,500 feet and a Class D Control Area (CTA) to the east between 2,000 feet and 3,500 feet.

There are also two Class D Control Areas (CTA) to the north and south between 1,500 and 3,500 feet and below these are TMZ's from the surface to 1,500 feet.

Above all of these upwards from 3,500 feet there is the London TMA, which is Class A.



5. Transponder Mandatory Zones. These are Zones within which aircraft must be equipped with an Elementary Mode S transponder. Some TMZs are coincident with controlled airspace, such as the Control Zone around London Heathrow, in these cases you must obtain an ATC clearance to enter as usual. Other TMZs are established outside controlled airspace, for example the TMZs below London Stanstead's control areas. Aircraft equipped with at least an Elementary Mode S transponder do not need to speak to ATC to enter, although it is recommended that pilots do or to use the appropriate monitoring code. Pilots of aircraft that are not suitably equipped must call the nominated ATC Unit to obtain approval to enter a TMZ established outside controlled airspace.

TMZs are marked on the chart by a series of magenta "half-moon" symbols facing into the TMZ. You may also encounter RADIO MANDATORY ZONES (RMZs), within which aircraft must be equipped with a VHF radio. These are marked on the chart with blue "half-moon" symbols

6. It is the Frequency Monitoring Code (FMC) for Stansted. They are also referred to as "listening squawks".

This facility exists for aircraft routing close to certain airfields but intending to remain outside controlled airspace. Selection of the appropriate code advises ATC that a pilot is monitoring the associated frequency. In this case Stansted's listening squawk 7013 is shown on the chart next to the frequency 120.625.

Should ATC see that an aircraft has, or is likely to, infringe controlled airspace the controller is able to call the pilot and advise them. The aim is to both prevent, and mitigate the effect of, airspace infringements. A full list of the frequency monitoring codes may be found in the AIP ENR 1.6.

7. The symbol represents a HIRTA, a High Intensity Radio Transmission Area. This is an area of defined dimensions within which there is intense radio energy which may cause interference or damage to an aircraft's radio equipment. These areas should be avoided.
8. The advice in the UKAIP states that when requesting penetration of a MATZ if no reply is received to two consecutive calls proceed with caution. Avoiding the area by turning or climbing above 3,032 feet AAL to fly above the MATZ are also good options!

ENR 2.2 OTHER REGULATED AIRSPACE

2.3.1 A MATZ Penetration Service will be available during the published hours of watch of the respective ATS Units. However, as many units are often open for flying outside normal operating hours, pilots should call for the penetration service irrespective of the hours of watch published. If, outside normal operating hours, no reply is received after two consecutive calls, pilots are advised to proceed with caution.

9. Moleworth is a Government heliport. (If required see list with practice route 3).
10. It is a Danger Area numbered 206 from the surface to altitude 6,000 feet.



END OF ANSWERS TO ROUTE 4 – PETERBOROUGH TO CONINGTON

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